

THE FACTORY, EXPLAINED SIMPLY

No code, no jargon. Just a cooking show that runs itself.

Forget code for a minute. Picture a **cooking competition that runs itself**: a coach picks which recipe ideas to try, cooks make the dishes, a machine tastes them, a critic scores them, and a writer publishes the cookbook — round after round, until a clear winner emerges. That is exactly what the factory does. The “recipes” are small neural-network designs and the “tasting” is a training run. Meet the cast.

1 · THE CAST

THE SHOWRUNNER

The one program in charge. It runs the schedule, keeps the official scorebook, and does every bit of paperwork — nobody else is allowed to write anything down.

THE COACH

A strategist. Before each round, he studies every past score and picks which recipe ideas are worth trying next.

THE COOK

A builder. Given one recipe idea from the coach, he actually cooks the dish.

THE TASTING MACHINE

Not a person at all. It tastes the dish and prints an objective score — same dish, same score, every time. No moods, no favorites.

THE CRITIC

A scorer with opinions. He rates each dish from 0 to 100, but must stay within 3 points of the machine’s score. He can nudge, never invent.

THE WRITER

The storyteller. After each round she writes a short chapter about every dish for the cookbook the show is publishing.

2 · ONE ROUND, START TO FINISH

1 PLAN

COACH reviews everything tried so far and proposes a few recipe ideas for this round — usually two to four.

2 COOK & TASTE

All dishes are made at the same time, and each one goes through the same little line: **COOK cooks it** → **MACHINE tastes it** → **CRITIC scores it**.

3 RECORD

The showrunner writes every score into the official scorebook, in order. A dish that beats the all-time best becomes the new **champion** — its recipe card goes on the wall.

4 WRITE UP

WRITER adds one short chapter per dish to the cookbook, so the story of the search keeps growing.

5 DECIDE

The showrunner asks: **are we done?** (The three finish lines are on page 2.) If not, the next round begins — and the coach plans it knowing every score that came before.

A ROUND UP CLOSE — AND HOW IT ENDS

Same show, closer look: one real-feeling round, the mishaps, the finish lines, and the one rule behind it all.

3 · A SAMPLE ROUND

Round 7. The coach notices wider cakes have been scoring well lately, so this round’s two ideas are: “**the classic cake, exactly as before**” and “**a wider version of it**”.

The cook bakes both. The machine tastes them: **71.2** and **78.5**.

The critic, staying within his 3 points, scores them **70** and **79**.

79 beats the old record of 76 — the wide cake becomes the **new champion**, and its recipe card goes on the wall.

The writer adds two chapters. The showrunner checks the finish lines: none met. **Round 8 begins** — and the coach now knows that wider cakes work.

4 · WHEN THINGS GO WRONG

- **A fumble.** If a performer returns a nonsensical answer or a broken dish, the showrunner simply asks again. Keep failing, and a more experienced stand-in finishes that one task. Nothing else changes.
- **Three strikes.** If one dish fails three times in a row, it is disqualified: noted in the scorebook, written off politely in the cookbook, and the round carries on without it.
- **The phone line dies.** If the show cannot reach its performers at all, it pauses on the spot. Restart it later and it picks up exactly where it left off — nothing lost, nothing redone.

5 · THE THREE FINISH LINES — any one ends the show

- 1 **Round limit.** The maximum number of rounds is reached.
- 2 **Out of ideas.** The coach has not proposed a genuinely new kind of recipe in 10 rounds.
- 3 **Ceiling reached.** The champion’s score has not improved in 10 rounds.

THE ONE GOLDEN RULE

Performers never touch the paperwork. The coach, cook, critic and writer only answer questions. The showrunner checks every answer, throws out the bad ones, and writes everything down personally. That is why the show is trustworthy: run it twice and the same dishes get the same scores — the only unpredictable part is what the performers come up with.

6 · BACK TO REALITY — what everyone is called in the actual system

the showrunner	run_factory.py — the main program	one round	a wave
the coach	the Meta agent	one dish	an iteration
the cook	the Worker agent	the scorebook	factory_state.json
the tasting machine	the Simulator (no AI involved)	the champion’s card	best_config.json
the critic	the Judge agent	the cookbook	the research paper
the writer	the Doc agent	a disqualified dish	a “poisoned” iteration